Neural Bellman Operators
Author Response to the Advisory Referee

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1 Report, manuscript, and evidence boundary

We thank the advisory referee for identifying the central unresolved problem: an improvement in a global continuation residual need not improve the decision that the Bellman service delivers. We have continued the author revision of Neural Bellman Operators around the original recursive economic decision problem. The paper retains neural continuation acquisition and actor/critic construction as research objects, develops certificates for the original Bellman objective, and compares the complete decision service with conventional alternatives. Econometrica remains the intended destination.

This response addresses the ten blocking concerns, ten major requests, and ten minor comments in the latest persisted substantive report. That report is attached to review commit `9723c6cd2cc8b3e071422f3e522d4a843946db63`, report blob `33ba8b144e2f7816b19b036404341402d23a582f`, and reviewed author commit `f802fe64d91d9f5c1c8f86123b0fbe114e622279`. It is a repository-owner-commissioned advisory report, not an editorial decision of the Econometric Society. No subsequent report of the completed R110 manuscript was found among the 36 review branches.

The reliable complete author baseline for the present revision is R110, commit `4ba2bbaa11b5fa59f83705b28b35bf82a63ca169`. Its ECTA master source has Git blob `fe8d082b96bb80834fdcfd5efc6675f5557e2efd`. Later numbered branches contain useful research work with distinct source and experimental commitments. Their larger numbers do not by themselves establish a more complete manuscript. In particular, a truncated R118 delta transport is not a complete replacement for R110. The revision boundary preserves those episodes and separates recovery, unchanged-source reproduction, new mathematical work, and new experiments.

Throughout this response, "retained" identifies results already present in the completed R110 source or earlier evidence. "Added" identifies mathematical results in the current author revision. An inherited execution receipt is evidence for its named source and campaign. It does not become an R119 execution receipt by being included in this package.

The current host has completed attributable reproductions of the unchanged frozen R117 and R118 executable episodes and three separately frozen R119 development phases. Each has its own source and execution identity. The mathematical additions received independent author-side proof and implementation inspection; exact record and numerical audits are reported separately below. The publication section records the actually completed native diagnostic build, remote custody verification and branch-preservation audit. Final source/PDF and remote workflow identities are bound outside the PDFs, so an output does not attempt to certify its own hash.

2 The substantive addition

The new information algorithm resolves the adverse part of the original innovation distribution before investing in fine quadrature. For a convex K-Lipschitz continuation on the original uniform interval [-R,R], write $t = 1 - \alpha$ and $L = 2R\alpha$. The complement of a central interval of length L is an admissible AVaR tail with total length 2Rt. The central integral H(u) has monotone, 2K-Lipschitz derivative

$$D(u) = f(u + L) - f(u).$$

After m paid sign bisections, a feasible selected tail has risk deficit at most $KRt\,4^{-m}/2$. Midpoint and secant information on at most n+1 cells within those selected tails encloses their risk average to width KRt/n^2. Thus the complete exact-point information width is at most

$$KR(1-\alpha)\left(n^{-2} + \tfrac{1}{2}4^{-m}\right), \qquad N \le 2n + 2\lceil \log_2 n \rceil + 8$$

when $m = \lceil \log_2 n \rceil$. Together with the retained lower width $KR(1-\alpha)/[6(N+1)^2]$, this establishes an error-versus-information order $KR(1-\alpha)N^{-2}$, with constants uniform in $\alpha \in [0, 1)$, for $N \ge 32$. The method pays for localization. It uses exact point values of a univariate convex Lipschitz continuation and does not assume access to its formula, derivatives, or integrals. The $\alpha = 0$ case uses ordinary whole-interval quadrature; $\alpha = 1$ is excluded from the displayed normalized-tail formula.

The original investment implementation supplies the missing connection between information and a realized decision. Its primitive curvature gives the innovation continuation the strong-convexity modulus 181/64. An outward interval for D(u), including an unresolved sign, produces a constrained strong-convexity defect κ(u). A feasible fixed tail and global original-continuation supports then enclose the cost of the actual proposed action and give a lower bound on the original optimum over the entire feasible action interval. The deployed terminal actor remains in the policy cost through its established allowance $\varepsilon_1 = 2^{-20}$. A new quadrature certificate never silently replaces that actor.

The supplement also develops common-tail action contrasts, an atom-safe remainder for AVaR, and signed recursive policy gains. These statements compare the original actions under the same conditional risk operator. They preserve the reference dual penalty and can discard reference densities whose slack exceeds the possible oscillation of the continuation gain. A paid precision allocation result balances the quadrature and tail-localization allowances under an explicitly stated work model.

These are information and decision-service contributions shared by every representation with the same information and feasible actions. Their value for learned NBO proposals must be assessed in the matched complete-work comparison. They do not constitute a neural-exclusive representation theorem.

2.1 A bounded, original-policy preliminary service

Proposition *bounded original-policy preliminary certificate* (11) makes the new precision choice executable. A floating endpoint comparison and at most six interior bisections propose a feasible central-window cut. They do not certify its sign or optimality. The two outward original-continuation endpoint values are purchased once, shared between attempted resolutions, and give the constrained strong-convexity defect used in Theorem *original-policy qualification from a feasible*

active tail (10). Thus a wrong floating proposal may be inefficient but cannot authorize an invalid decision.

Only tails of exactly positive dyadic length receive midpoint and endpoint payloads. With $p \in \{1, 2\}$ positive tails, the two preliminary resolutions $m = 2, 4$ purchase at most

$$2 + p\{(2 \cdot 2 + 1) + (2 \cdot 4 + 1)\} = 2 + 14p \leq 30$$

outward original-continuation values. A boundary tail needs at most 16. The separate floating proposal count is at most $4 + 2 \cdot 6 = 16$. These are additional outer payload caps per proposed action, including the two defect endpoints. Internal compiler work, duplicate requests, model fitting, rejected attempts, original 128-bin fallback, further conventional repair and any eventual exact recovery remain charged separately. The positive-tail masked ablation instead retains the full six-level ladder, whose corresponding outward preliminary cap is $2 + 258p \leq 518$.

The packed implementation retains that same two-level policy. It concatenates the positive-tail midpoint and endpoint coordinates with their original state and action owners, evaluates them in one original compiler call per attempted resolution, and scatters the resulting interval values and slopes back to the same columns. An exact zero tail contributes zero mass; outward arithmetic still encloses every zero product and sum. The mathematical statement is reindexing and sound enclosure, not bitwise identity: batch-dependent compiler fallback can return different valid bounds. Complete returned-policy identity must therefore be checked in the actual records. The source and full proof are `new119/budgeted_tail_main119.tex`, `new119/budgeted_tail_proofs119.tex` and the packing statement in the supplement. Neither payload cap nor packing proves a wall-clock saving.

2.2 Bellman foundations, constrained gradients and stopping boundaries

The new foundations supplement states sufficient assumptions instead of inferring well-posedness from a residual loss. On compact state spaces with continuous compact feasible-action correspondences, continuous bounded primitives and terminal values, the conditional risk operator must preserve parameter measurability and continuity under essential-uniform changes of continuation. Fixed-law expectation and AVaR satisfy the stated requirements; the AVaR measurability proof takes a countable infimum over rational thresholds. Backward induction then gives continuous finite-horizon values and Borel minimizing selectors, and a separate policy argument compares them with all admissible history policies under the same nested criterion. Stability on this Bellman space is proved for bounded continuous terminal values. A fixed Borel policy has its own bounded-Borel comparison under the kernel property. These different continuation classes are not silently interchanged.

For the original investment model the proof derives the invariant state cube, joint state/action curvature and unique feasible minimizers from the nonlinear primitives. It establishes the original value and action-objective differentiability needed by the certificate, including the constrained envelope

$$\nabla V(x) = Q_x(x, a^*(x)) - \lambda(x) \nabla A(x).$$

The capacity term cannot be discarded at an active upper action boundary. The actor itself need not be globally differentiable across changes in the active constraint. Downward 40-bit state observation has a separate feasibility and policy-loss transfer; it does not claim that an arbitrary discontinuous rounded policy can be differentiated as the exact optimizer.

For the actor/critic graph the stated objective is $L(\theta) = \frac{1}{2}\mathbb{E}[r_\theta^2]$. Under the declared differentiability and domination conditions its derivative includes the current critic path, the explicit future-parameter path and the actor path. Detaching a target differentiates a different surrogate. The AVaR derivative needs its stated unique, atomless quantile condition; neither graph identity establishes optimizer convergence. The original strong convexity makes that risk regularity check possible here, rather than assuming every neural continuation has it.

The stopped Brownian BSDE result uses bounded $C^{1,2}$ test functions, stated coefficient and driver regularity, the actual stopping payoff and a well-posed closed-loop feedback. Boundary discrepancy and both signed residual errors enter the policy-gap bound. A zero interior residual alone can leave positive stopping-policy regret. A ReLU function's zero Hessian almost everywhere is therefore not a classical Ito verification argument without a justified weak formulation or smoothing error. General NDU, fully coupled FBSDEs, temporal-self equilibria and games retain their distinct mathematical obligations. These corrections are proved in `new119/foundations_supplement.tex`, with the main statements in `new119/foundations_main.tex` and the learning graph in `new119/policy_main119.tex`.

3 Completed reproductions with preserved episode identities

The R119 host executed the unchanged frozen R117 and R118 sources under custody commit `54512c9c7fcdc75d97ad37d20b8b61f22356a953`. These executions preceded and informed the new R119 design. They are attributable reproductions, not previously unseen R119 active-tail observations and not recovered observations from an earlier host.

3.1 R117: complete original-investment reproduction

The original R117 executable freeze is `a21e33471047b966dd375d0b9aceab6b9492e4fb`, with protocol commit `e65a0180896406610922e48f65bcf7f55ada366f`. The current reproduction completed 4,608 original-investment primary decisions and 7,168 timing decisions in seven sequential fresh processes. Both preflights, the full campaign, the frozen record audit, and the additional independent exact-diagnostic audit returned exit code zero. Their seven combined R117/R118 command logs were independently checked against the receipt hashes.

The R117 audit checks 11,776 decision rows, 8,007 paid primary precision-attempt rows, 204,912 exact-rational record inequalities, and 3,072 independent primitive checks. ReLU and its conventional hinge encoding return the same 144 actions and certificates for each risk/service pair. Across all 2,304 matched primary policies, geometric and fixed-128-bin services return identical actions and the same future controller. Their economic policy-cost contrasts are therefore exactly zero.

The computational finding is mixed. Across the primary comparison, the recorded continuation locations decrease from 654,836 to 223,767, while the 128-bin repair-gate rejection count remains 244 for both services. The observed 32-decision median clock is larger under geometric scheduling in five of the sixteen risk/representation pairs. In particular, the ReLU medians increase from 0.086081 to 0.091844 seconds at $\alpha = 1/2$ and from 0.088763 to 0.096323 seconds at $\alpha = 3/4$. The corresponding ordinary hinge medians also increase. These are descriptive clocks from this reproduction, not timing confidence intervals or replacements for older clocks.

With the audit's full standalone acquisition charge for the declared 32-decision workload, the conventional online geometric service has the smallest recorded total at both risk levels: approximately 0.073497 and 0.074537 seconds. The complete all-arm table below is retained to avoid

selecting only the faster cells. This accounting is conditional on that workload and observed host; it does not assign an external economic compute price.

Risk	Representation	Fixed 128-bin median, s	Geometric median, s	Median ratio
1/2	online	0.100386	0.073435	0.7315
1/2	quadratic	0.057446	0.033314	0.5799
1/2	relu	0.086081	0.091844	1.0670
1/2	hinge	0.079147	0.086764	1.0962
1/2	spline	0.056986	0.031767	0.5575
1/2	poly2	0.220188	0.218858	0.9940
1/2	poly3	0.269224	0.247821	0.9205
1/2	grid	0.221658	0.198796	0.8969
3/4	online	0.098671	0.074489	0.7549
3/4	quadratic	0.057397	0.032429	0.5650
3/4	relu	0.088763	0.096323	1.0852
3/4	hinge	0.078779	0.086200	1.0942
3/4	spline	0.057797	0.033024	0.5714
3/4	poly2	0.216773	0.188198	0.8682
3/4	poly3	0.265188	0.255670	0.9641
3/4	grid	0.214650	0.215377	1.0034

The independently reconstructed exact supporting R117 diagnostic checks 120 method rows, 459 precision attempts, and 5,116 rational identities or inequalities. The action-gap rule qualifies all 60 of its cases using 2,484 point requests and 1,484 unique queries. The direct risk-width rule qualifies 51 of 60 and leaves nine unresolved, using 245,431 requests and 122,844 unique queries. This exact supporting information problem is separate from the original-investment campaign. Its favorable action-gap account does not turn the unchanged investment policies into economic improvements.

The executed records are `results119/reproduction117/REPRODUCTION119.json`, `AUDIT117.json`, and EXACT 117_`INDEPENDENT119.json`. The reproduction receipt has SHA-256 `a4d421ad5adb817756c64876d18d4b7efc07124df8daa236dc4b4a37fca53c7a`; the independent exact audit has SHA-256 `f49699f492f742595cbc40171bd69b43cc66411c2783e0cc77b83ecf669c9ce4`. The original R117 remote recovery failure remains a failure; this new-host reproduction does not relabel it.

3.2 R118: exact scalar decision-recovery reproduction

The recovered R118 executable is intact even though its transported article delta is incomplete. Its original executable freeze is `17ed16688f2f66924915192d3db7035c86bc846c` and its protocol commit is `7f6864e9fca9929b4f171e44ce3c558614b8ea06`. The supporting model is the declared quadratic Bellman oracle subproblem

$$Q_\theta(a) = \frac{\mu}{2}(a - a_0)^2 + \beta a\theta, \qquad a \in [0,1], \quad \theta \in [l,u].$$

Using the same supplied parameter interval, its minimax-regret action has strictly smaller worst-case regret than the projected midpoint-parameter policy in 352 of the 792 fixed cells, and

equal regret in 440. There are 68 cells with nonzero parameter width and zero regret. All cells, including zero-width and shared-boundary cases, remain in the catalogue.

The independent audit performs 41,403 exact record checks and enumerates all 26,136 prescribed grid actions. Production uses 7,240 endpoint objective evaluations; the separate reference account uses 70,134. Both executed commands return zero. The audit's SHA-256 is `02aa86306b8ae899abc27cd853e2c4dac58341a805f65b2394e839824ebf5178`, and the reproduction receipt's SHA-256 is `52725ae8a44ff9fdc100008b1893cf5663d4a41239c0428fb72c4ff9963768dc`.

This completes a reproduction of the frozen exact diagnostic. It does not complete every theorem proposed in the R118 protocol, replace the original nonlinear investment model, supply R115 population data, or establish a neural representation advantage. The separate scientific contribution is that interval information can be sufficient to choose an optimal constrained decision even when that interval retains positive width.

4 Blocking concerns

4.1 B1. Neural specificity remains unestablished

Response. We agree that representation-neutral validity cannot establish a neural-specific advantage. The revised operator construction states precisely which objects are learned: a continuation representation, its associated proposal, and any acquired certificate information. The original economic loss, feasible actions, original conditional risk operator, terminal policy, and acceptance predicate remain common to the neural and conventional services.

R110 already makes the ReLU/min-plus and ordinary oblique-hinge equivalences explicit. The new active-tail algorithm is supplied to the same conventional representations. No geometric feature available to a hinge, triangulation, or polynomial supplied with the same information is counted as uniquely neural. A within-ReLU improvement over an older service is not the frontier comparison.

Locations and status. Retained: `new110/core-main.tex`, `new/decision99.tex`, and `new110/study.tex`. Added: Theorems *active-tail information with uniform risk-level constants* (9) and *original-policy qualification from a feasible active tail* (10). Neural-exclusive complete-work superiority or certified adoption over the strongest conventional service remains an open empirical obligation. The current R119 experimental record below reports the current all-arm frontier and its complete charged costs.

Completed R119 action and evidence. All three phases now execute the shared service for every arm. Phase C's packed service benefits the ordinary hinge as well as ReLU. Quadratic remains cheaper for the full declared workload and has the lower catalogue-mean original economic cost at both risks. The representation-neutral contribution is therefore identified separately from the still-unestablished neural competitive increment.

4.2 B2. The information-complexity result is too narrow

Response. The former whole-support upper enclosure paid an unfavorable tail factor while the lower bound decreased with tail length. The new theorem removes that mismatch under the same exact-point univariate uniform-law information convention. It first locates a risk-active pair of endpoint intervals and then acquires continuation values only there. The localization cost is explicitly counted, and the upper and lower widths both scale linearly in $1-\alpha$. This is a stronger statement than exponent agreement at each fixed α.

The proof requires convexity and a known Lipschitz bound. Its finite-budget statement uses $N \geq 32$. Exact-point information does not include the bit cost of resolving nearby endpoint values as α approaches one. The original investment implementation separately accounts for uncertain values and outward arithmetic. R110's nonuniform-density result, pathwise-sound randomized-width statement, ordinary randomized lower bound, and dated precision composition remain available with their own quantifiers.

Locations and status. Added: Theorem *active-tail information with uniform risk-level constants* (9) and its complete proof. Retained: `new/information99.tex` and `new/precision99-link.tex`. Optimal general randomized rates, a comparable multivariate information theorem, and a full learned end-to-end Bellman complexity result remain open. The new exact-point theorem does not claim these results.

Completed R119 action and evidence. The completed exact supporting catalogue contains all 320 cells and 2901 rational checks, including the $\alpha = 0$ case and risk levels through $255/256$. It preserves 242 active-tail width wins,27 whole-support wins and 51 ties at the same declared query cap. These examples validate implementation in their fixed class; the uniform N^{-2} theorem follows from the full proof, not these counts.

4.3 B3. The representation result is not decision-theoretic

Response. The completed R110 source already embeds the oblique continuation in an exactly specified two-date posted-price Bellman problem. Under its declared price law, the expected policy regret is the squared continuation error divided by six. The exact regrets are $1/540$ for the best additive continuation, $1/5400$ for the best total-degree-two polynomial, and zero for both the exact ReLU representation and the equivalent conventional hinge or triangulation. This establishes a policy mechanism rather than an approximation norm alone.

The posted-price economy remains a distinct supporting example with formula access. We do not present its exact regret as a performance improvement in the original investment experiment or as evidence that a neural learner discovers the oblique direction more efficiently. The new original-investment theorem complements that example by turning original-continuation information into a certified loss for a realized original action.

Locations and status. Retained: `new/decision99.tex`. Added: Theorem *original-policy qualification from a feasible active tail* (10). The exact supporting example answers the missing policy connection under its stated economy. Unknown-geometry acquisition and neural-exclusive computational advantage remain unestablished.

4.4 B4. The certificate-aware objective fails its service connection

Response. The original adverse result remains central. In R97, all 36 certificate-aware fits reduce their validated residual span, while total first-proposal rejections increase from 644 to 757. We do not relabel that fit as successful decision-focused learning.

The current mathematical object is an original-action gap. A feasible original-risk witness and valid global continuation supports give a lower quadratic bound for every feasible action; an outward policy upper endpoint then bounds regret relative to the original optimum. The active-tail algorithm reduces a component of the information needed to resolve that gap. Its κ term charges errors in selecting the adverse tail, including the possibility that an interval derivative has unresolved sign. The common-tail comparison additionally bounds an actual policy-cost difference, with the correct one-sided terminal allowance enlarged to $\pm\beta\varepsilon_1$ for a contrast.

Locations and status. Retained: `new110/action-main.tex`, `new109/tail.tex`, `new109/improvement.tex`, and `new110/budget.tex`. Added: Theorems *original-policy qualification from a feasible active tail* (10) and *atom-safe coupled action certificate* (5). These are substantive certificate and service changes. They do not by themselves constitute prospectively validated action-aware fitting. The current R119 experimental record below reports whether the new information allocation improves complete work for the frozen original-economy proposals.

Completed R119 action and evidence. The bounded prelude and packed implementation now use the original-action certificate as the stopping predicate. All three phases are complete, and every within-arm service contrast is exactly zero because the complete policies coincide. The result improves parts of information acquisition and serving cost; it does not turn the unchanged least-squares acquisition objective into validated action-aware learning.

4.5 B5. Complete work favors a strong conventional method

Response. The quadratic least-squares winner remains part of the evidence. R110 broadens the frontier and makes the original-model semi-analytic compiler available to every arm. The compiler proposes terminal solutions under the same primitives; an original outward witness, rather than an unverified algebraic root, grants validity.

The new active-tail service uses that same common information interface. Its cost account must include localization queries, training and model selection if any, certificate construction, unresolved signs, rejected certificates, refinement, fallback, and actual inference. An operation reduction on accepted ReLU proposals would not establish superiority over a conventional service that incurs fewer proposals or lower acquisition cost.

Locations and status. Retained: `new109/compiler.tex` and `new110/study.tex`. Added: active-tail service and paid-precision account. The prior conventional winner is not overturned by a theorem about a shared algorithm. The current R119 experimental record below reports the matched frontier, including every conventional arm that qualifies under the same loss and width predicates.

Completed R119 action and evidence. Phase A loses against geometric qualification in 15/16 median-clock cells; phase B's two-level budget still loses in 14/16. Phase C's packed version wins 10/16 against its fresh geometric control and 16/16 against its fresh budgeted and compiled controls. All six geometric losses and all conventional acquisition costs remain reported. The quadratic overall frontier is not overturned.

4.6 B6. The purchase exercise is conditional sensitivity

Response. We retain the distinction between a unit-consistent purchase identity and an identified adoption decision. The R97 grid contains 156 adoption, 199 rejection, and 1,445 unresolved classifications among 1,800 conditional cells. Every certified adoption is relative to the online solver; none is over quadratic least squares.

The original Bellman policy loss is now linked more directly to information expenditure. The remaining-gain stopping result bounds the economic improvement deliberately forgone; the new action certificate and coupled contrast can narrow numerical policy-cost intervals. These improvements do not calibrate deployment prices, fixed fees, workload persistence, or an institutional welfare margin.

Locations and status. Retained: purchase account, `new110/budget.tex`, `new110/study.tex`, and the exact supporting posted-price model. Added: coupled original-policy contrast.

A consequential calibrated application remains open. Conditional sensitivity and unresolved adoption regions stay labeled as such.

4.7 B7. The newly fitted actors require their own population inference

Response. We agree. The R96 population intervals apply to their own frozen actors and sample. They cannot establish the mean-cost ordering of the R97 actors, R110 refits, or a new controller. The completed R110 experiment is a deterministic whole-grid catalogue; its 64 evaluation states are not represented as an independent random sample from a newly specified population.

The revision boundary separately records source freezes, actor identities, data commitments, and execution status for later episodes. Recovery or reproduction of one episode does not complete a different frozen randomized campaign. New certificate nodes can concern the same fixed realized policy, but any changed deployed action or terminal controller requires a correctly identified policy estimand.

Locations and status. Retained: `PROTOCOL110.md`, `new110/study.tex`, `new/precision99-link.tex`, and `REVISION_BOUNDARY119.md`. Current: revision boundary and source-binding ledger. New-actor population policy-cost inference remains open unless a current source-bound independent sample and its prescribed inferential procedure are actually executed. No R96 confidence interval is transferred.

4.8 B8. Independent verification stops before fitting

Response. R110 already adds a current acquisition reconstruction: independently assembled features, separate active-set enumeration, pivoted QR, agreement of selected rounded coefficients, a separate polynomial factorization, and serialization/reload checks. The shared floating-point libraries are disclosed. This provides an independent route through the current fitting calculation, but does not retroactively reproduce the R97 interval-aware optimizer.

For the new active-tail method, mathematical independence also matters. The exact-point rate was checked from the central-window geometry, its derivative, the two-tail secant variation, and a conservative query count. The original investment theorem was checked separately from primitive curvature and global supports. Such checking is not a substitute for executing an independent numerical implementation.

Locations and status. Retained: current acquisition reconstruction and source-support110 closure. Added: complete active-tail and coupled proofs; author-side independent check. The current R119 experimental record below identifies the independent numerical reconstruction and receipt for the new implementation. Historical replay errors and the historical failed assertion remain separate adverse results.

Completed R119 action and evidence. Each phase's original-primitive validator completes 22220 checks, including 72 original policy certificates, under its own frozen source. Exact record audits separately check 444053,1031933 and 1325935 conditions in A, B and C. The C packing reconstruction verifies 21974 additional identities. These are actual, bounded scopes of independence; shared numerical libraries and reused deterministic test contexts remain disclosed.

4.9 B9. The active submission remains cumulative and disproportionate

Response. The scientific sequence of the main article is the original economic Bellman problem, acquired continuation and proposal, action-relevant certificate, active-tail information algorithm, original-policy improvement and stopping, and the matched numerical comparison. The retained

exact supporting model and delimited stochastic-control bridge explain specific mathematical connections. Historical development detail belongs in the companion.

A content-retention map preserves theorem, proof, code, evidence, and response lineage when material moves. A historical result that was disproved or whose supporting curve was not independently generated is retained as a corrected development record. Its archival presence does not restore it as an active theorem or empirical finding.

Locations and status. Retained: ECTA.tex and the new110 main/archive split. Current: content-retention map, supplement, and development companion. The current publication record below identifies the actual built artifacts, source/PDF hashes, page counts, and completed checks; source organization alone is not a verified artifact.

4.10 B10. The negative result should generate an action-aligned contribution

Response. We make the certificate-to-decision mismatch a productive scientific question within Neural Bellman Operators. The reason a global residual span can fail is that its extrema and scale need not identify the adverse continuation outcomes or action directions that determine acceptance. The current method uses the original active risk tails, a constrained action support, and a complete-policy contrast. The mathematical target is the information needed to settle a feasible decision at the original loss and width thresholds.

The new upper information result and original-investment implementation provide a concrete algorithmic response. The signed recursive gain equation further retains the dual slack of the reference policy, so a small possible policy change does not automatically incur the whole worst-case adverse-law envelope. These results continue the original research program; they do not reinterpret the adverse span experiment as having accomplished its intended training objective.

Locations and status. Added: active-tail and coupled sections. Retained: negative R97 catalogue, `new109/tail.tex`, `new109/improvement.tex`, and `new110/budget.tex`. Prospectively validated action-aware acquisition, stronger complete-work competition, and substantive adoption remain active research obligations.

Completed R119 action and evidence. The scientific iteration is explicit: A exposed tail-schedule overhead; B bounded extra information but still lost most geometric clocks; C packed exactly the same positive-mass payloads and improved the contemporaneous complete service in ten median cells. Theorem-level information gains and observed implementation gains are kept separate from exact-zero policy gains and the continuing action-aware acquisition problem.

5 Major changes

5.1 M1. Recast the paper around one method-neutral question

Response. We retain Neural Bellman Operators as the central research identity and state the common decision question precisely: how should a learned continuation and actor use information to produce a feasible decision for the original recursive economic objective at justified total cost? Neural representation is a candidate source of computational benefit. The validity of the original-risk certificate is established separately and is shared with conventional representations.

The main article follows that question through an explicit Bellman operator, a stored continuation, its realized actor, original-policy cost, and matched service comparison. The revised contribution does not depend on attributing a common hinge geometry or a valid fallback to neural networks alone. The report's scientific concern is addressed inside the original paper; the author's

decision to continue the NBO/Econometrica program does not count as a mathematical answer to the concern.

5.2 M2. Establish the novelty of the information theorem

Response. The new theorem distinguishes three ingredients. The AVaR variational representation, convex midpoint/secant inequalities, and indistinguishable-query lower-bound method are established tools. R110's nonuniform and randomized statements are retained with their own scope. The current addition uses the geometry of a convex continuation to locate the adverse tails and then allocates point information within their shrinking support, with a paid localization count.

The result closes the previous difference in tail dependence: both deterministic lower and constructive upper widths are proportional to $KR(1-\alpha)N^{-2}$, uniformly in $\alpha \in [0,1)$ for $N \geq 32$. The original-investment theorem then supplies an outward implementation of that geometry with uncertain endpoint values and a certified original-policy gap. It does not equate exact formulas with value queries or claim an optimal general randomized or multivariate rate.

Location. Theorem *active-tail information with uniform risk-level constants* (9), original-policy Theorem *original-policy qualification from a feasible active tail* (10), their proofs, and the novelty/source table. The supplement credits the classical AVaR, convex-integration, and information-indistinguishability ingredients at their first use and states the complete new proof separately.

5.3 M3. Develop an action-aligned acquisition objective

Response. The mathematical target is an original-action upper regret bound, not a global residual span. The lower side is the minimum of an all-action supporting quadratic under a fixed feasible original-risk witness. The upper side is the cost of the realized proposed action, including the deployed terminal controller. The new localization defect and quadrature width identify which information components can keep that action unresolved.

A finite resolution ladder accepts only when the unchanged original loss and width predicates hold. Every earlier unsuccessful attempt remains charged. The paid allocation calculation balances A/n^2 against $B4^{-m}$ for declared positive costs and checks $n \geq 1$ and $m \geq 0$ boundaries; it is not a global discovery optimum. The common-tail contrast can support a strict improvement over a completed incumbent when its upper difference endpoint is negative.

Status. These supply implementable certificate targets and conditional work allocation. They do not certify that frozen least-squares acquisition is now action-aware training. A prospectively frozen fitting-to-service experiment is required to close the acquisition request itself, with equal information, rank restrictions, feasible policies, validation budget, and charged rejected candidates. The current R119 protocol keeps inherited least-squares fitting unchanged; its certificate intervention does not complete that separate acquisition experiment.

Completed R119 action and evidence. The added bounded preliminary service supplies an implementable stop or fallback rule with 30 outward and 16 floating payload caps. Its soundness uses the original feasible-action support even when a floating cut is inaccurate. No new fitting criterion was selected or retrospectively described as preregistered; action-aware continuation acquisition remains the next unresolved mechanism.

5.4 M4. Expand the conventional geometry frontier

Response. The retained comparison includes ordinary oblique hinges with the same features as ReLU, coordinate splines, full quadratic and cubic alternatives, a bilinear grid, and an original-

model semi-analytic terminal compiler. The exact supporting decision model includes the conventional triangulation that represents the same hinge exactly. Differences in rank and storage between restricted and full polynomial families are explicit.

The active-tail and coupled certificates are available to all these families. A shared feature, a shared tail search, or a shared compiler cannot identify a neural representation effect. Adaptive triangulation, convex regression, and automatic basis discovery also consume acquisition and validation work and must be compared at that complete boundary.

Status. The frontier is broader than R97, but a completed equal-budget acquisition tournament over all the requested adaptive conventional classes remains open. The current R119 experimental record below gives the all-arm counts and costs for its actual fixed comparison; no inference is drawn from an unexecuted adaptive comparator.

5.5 M5. Treat the negative result as first-class evidence

Response. The R97 result is retained with its paired cell ledger: 36 of 36 lower spans; 11 cells with fewer first-proposal rejections, five unchanged, and 20 with more. The aggregate rises from 644 to 757. The family counts are quadratic 91 to 210, ReLU 286 to 287, and spline 267 to 260. The quadratic least-squares complete-work winner and zero learned adoption over it remain visible.

The new theorem explains a different acquisition target without changing those outcomes. Any current improvement belongs to its own source, algorithm, and fixed evaluation protocol. An inherited negative clock or rejected candidate is not replaced by a more favorable rerun. The development companion also preserves the earlier FVI wins, pure-ReLU adverse comparisons, and vector workload failures.

Status. Addressed by the retained content and its source ledger; package inclusion and hashes must be checked at release.

5.6 M6. Execute a consequential application

Response. The original investment model continues to define the numerical decision problem. Its reported objectives, regret enclosures, and foregone-improvement accounts have economic units. The supporting posted-price model supplies an exact mechanism showing how a continuation representation changes a policy, under its own separate primitives.

The new active-tail certificate can reduce uncertainty about the original policy cost, and the coupled recursion can preserve a signed gain instead of immediately replacing it by an unsigned norm bound. Neither result calibrates an external purchase decision. We retain the goal of a substantive economic application and identify the required evidence: credible workload and compute prices, policy margins under the original model, and a comparison with the best feasible conventional service.

Status. An empirically or institutionally calibrated adoption result remains open. The conditional grid and the separate exact example do not close it.

5.7 M7. Add a genuine scaling experiment

Response. The completed source distinguishes independent d=2,4,8 kernel checks from a learned tournament over horizon, control dimension, and innovation dimension. The latter requires new complete recursive-policy execution. The active-tail theorem concerns a single continuous innovation coordinate; it does not establish multivariate tractability.

At earlier dates, the witness construction can be used when the original continuation along that innovation retains the required convexity and provides global supports. This is a conditional theorem, not an assertion about an arbitrary learned network or an unexecuted longer-horizon controller. Product-point costs, descendant work, acquisition, memory, and failures remain necessary parts of any scaling comparison.

Status. Genuine learned multi-horizon, multi-control, and multi-innovation competition remains open. Historic adverse vector evidence is retained and is not replaced by a scalar diagnostic.

5.8 M8. Strengthen timing evidence

Response. R110 increases the descriptive timing campaign to seven sequential fresh-process repetitions, uses one computational thread, and retains raw clocks and operation categories. It explicitly discloses uncontrolled processor and background-load conditions. The current revision makes original terminal acquisitions, tail localization, endpoint work, failed attempts, fallback, and full service operations the primary portable workload quantities.

Clock ranges describe those actual runs. They are not confidence intervals for deployment latency, hardware-general speed guarantees, or empirical energy prices. New repetitions require their own identified source and prospective protocol; they cannot replace the original R97 timing sample.

Status. Retained timing improvement is source-specific. The current R119 experimental record below reports current hardware/process metadata, repetition receipts, raw clock ranges, and all charged work categories. No probabilistic timing inference is claimed without its specified sampling target.

Completed R119 action and evidence. All seven timing processes in every phase completed under that phase's fixed route order. The observed phase-A response-formatting overlap is recorded, and no optional retiming replaces those clocks. B and C use fresh contemporaneous controls after their source freezes. Tiny polynomial median wins in C are identified as weak descriptive evidence, with every paired clock and range retained.

5.9 M9. Independently reimplement fitting and package the environment

Response. The current acquisition reconstruction traverses data, features, constrained selection, rounded coefficients, evaluation, serialization, and reload. It discloses numerical-library sharing. The R110 source-support scheme and reproduction scripts provide a concrete dependency closure to reconstruct, rather than assuming that manuscript source alone is executable.

R119 must additionally bind the recovered executable modules to their source identities, preserve the unsuccessful truncated-transport attempt, and execute the new independent implementation from the packaged closure. Successful recovery of a legacy module does not establish the result of a frozen later experiment. Historical replay failures stay in their own ledger.

Status. The current R119 experimental record below reports the new source-bound independent numerical audit and executable dependency closure. The completed R117/R118 reproductions above provide actual acquisition-to-record evidence for their exact frozen episodes; they do not supply a current active-tail audit or eliminate the retained historical failures.

Completed R119 action and evidence. The A/B/C source freezes, preflights, acquisition audits, independent primitive checks, full attempt ledgers and read-only exact record checks are now available. The phase-B first transport attempt failed its byte-binding check before science because of CSV newline transport; the exact original bytes were restored, the failed attempt retained, and the algorithm and protocol unchanged. Current publication validation remains a separate receipt.

5.10 M10. Shorten the active submission

Response. The main text centers on the original economic question and the new action-information contribution. The full mathematical assumptions and proofs remain in the supplement. Historical source reconstruction, superseded diagnostics, prior negative studies, and report-by-report development belong in the development companion and machine-readable ledgers.

This organization preserves intellectual and empirical history without making every development episode a current main-text contribution. Content retention records whether each item is active, supporting, corrected, unresolved, or archival. A source hash proves identity, not scientific validity.

Status. The current publication record below identifies the actual built artifacts, source/PDF hashes, page counts, and completed checks; source organization alone is not a verified artifact.

6 Minor comments

6.1 Minor 1. Error-versus-information terminology

We use "error-versus-information rate" for the width order N^{-2}. The AVaR interval width, a Bellman action gap, a policy-regret rate, and an empirical operation count are different objects. The new theorem describes the first and states separately how its original-investment implementation contributes to the third.

6.2 Minor 2. Oracle qualifiers next to high-level claims

The exact-point theorem is explicitly deterministic, univariate, uniform-law, convex, and Lipschitz. Its constants are now uniform in the risk level on $\alpha \in [0, 1)$ for $N \geq 32$, so "fixed α" is no longer a sufficient description of this new upper/lower comparison. The retained nonuniform and randomized statements retain their own conventions. The abstract and conclusion must carry the relevant qualifiers wherever a rate is summarized.

6.3 Minor 3. Tail dependence of constants

The former opposite behavior of the constants is retained as the motivation for the new algorithm. Tail-localized quadrature and paid localization now produce a common linear factor $1-\alpha$ in the exact-point bounds. The theorem excludes $\alpha = 1$ from its normalized formula and handles $\alpha = 0$ separately. Finite-precision and outward-evaluation costs are explicitly outside the exact-real query count and are charged in the implemented service.

6.4 Minor 4. Coefficient restrictions and sensitivity

Coefficient signs, bounds, dictionaries, rank, selection rules, and rounding are acquisition restrictions. They do not characterize the unconstrained best approximation in the nominal representation family. Existing fitted coefficients remain tied to their frozen restrictions. A broad new coefficient-bound sensitivity study is not claimed; if run, it requires a separately identified protocol and complete acquisition cost.

6.5 Minor 5. Residual span versus action gap

The revised terminology distinguishes the validated global residual span, selected-tail localization defect, quadrature enclosure width, original-action supporting gap, complete-policy regret bound, and signed policy-cost contrast. Tables should keep their units and denominators separate. A smaller value in one column is not described as improvement in another without the corresponding theorem or executed comparison.

6.6 Minor 6. Rejections beside span improvements

Every summary of the R97 all-span-improved result is accompanied by the 644-to-757 rejection increase and the family decomposition. The complete paired cells and rejected paths remain in the supplement/data. Any new current table must likewise show accepted actions, rejections, fallback, and charged unsuccessful attempts beside certificate widths.

6.7 Minor 7. Observed acquisition range

Acquisition and service clock brackets are observed engineering ranges for named runs. They are not confidence intervals. Population policy-cost intervals, deterministic numerical enclosures, and descriptive timing ranges are labeled separately.

6.8 Minor 8. Zero adoption over quadratic least squares

The zero-adoption conclusion remains explicit for the retained R97 grid: all 156 certified adoption classifications are relative to the online service, and none is over quadratic least squares. A current shared-certificate improvement or separate diagnostic does not overwrite that historical result.

6.9 Minor 9. Verified approximation and HJB-residual literature

The novelty map distinguishes classical convex risk duality, midpoint/secant integration, information indistinguishability, approximate dynamic programming and residual propagation, and Brownian Markov BSDE verification from the actual new tail-information and action-coupling results. References belong beside the claims they support. The retained stochastic-control bridge states terminal/stopping boundaries, admissibility, and the graph relation $z = \sigma^{\mathsf{T}} \nabla u$; a residual penalty alone is not treated as a general viscosity or optimal-policy certificate.

The source comparison distinguishes those established arguments from the stated new theorem. A related-work paragraph does not substitute for explicit assumptions, the full proof, or evidence that a proposed application satisfies its hypotheses.

6.10 Minor 10. Title and neural-specific content

Neural Bellman Operators remains the title and center of the author revision. The text states that the operator is learned and used to form and compare continuation-based decisions; it does not imply that neural networks uniquely supply risk validity, a hinge geometry, or the new active-tail algorithm. A neural-specific empirical claim requires an equal-information comparison against representations with the same features and feasible policies, together with their complete costs.

7 Earlier reports that remain scientifically effective

The new foundations supplement now supplies an explicit compact finite-horizon proof of existence, continuity, Borel minimizing selectors, and domination of all admissible history policies under the specified nested criterion. The risk map must preserve parameter measurability; fixed-law expectation and AVaR are proved to do so. The same-primitives stability theorem is stated on its justified continuation class. For the original investment model, invariance, joint curvature, unique constrained actions, the capacity-aware value gradient, and the downward-observation terminal-policy transfer are proved from the primitives. The stopped BSDE result separately charges the actual stopping payoff and boundary error, and the actor/critic graph distinguishes the full derivative of the half-squared residual objective from a detached-target update. These additions respond to the early reports without extending their conclusions to an arbitrary NDU, a fully coupled FBSDE, or a general discontinuous feedback.

The report inventory contains 35 persisted documents across 36 review branches. It includes a threshold-screening document and should not be described as 35 independent substantive assessments. The R16 branch contains no newly persisted R16 report; its reviews subtree repeats R15. The R50 threshold screen explicitly declines a new substantive review. Four genuine earlier reports absent from the cumulative R97 reviews subtree—R14, R59, R61, and R67—have been recovered with their original blobs.

The exact document registry,620 explicit R/B/M comment blocks and 1146 hierarchical sections form the preservation ledger. The current `HISTORICAL_CARRYFORWARD119.json` supplies a comment-specific author treatment, concrete current proof/code/evidence locations and an explicit remaining boundary for every one of the 620 blocks. `ADDITIONAL_REFEREE_RESPONSE119.json` and its typeset companion add 68 explicit minor/additional/technical items,295 supplementary requirements and 31 embedded requests. These are 1014 routed records, not 1014 independent scientific issues. The 35 parent treatments overlap their 295 children, and the ten R97 minor comments also appear above. All 1146 overlapping hierarchical sections have an explicit classification; containers, historical findings, scope statements and editorial recommendations remain preserved. Original report text, report and commit identities, line bounds and comment hashes remain intact. No blanket scientific closure is assigned. The ledger does not mark a comment closed merely because it is old, repeated, included in an archive, or mentioned in a response. The current scientific treatment of the recurring requirements is as follows.

Historical requirement	Treatment in the current NBO article and supplement	Remaining boundary
Define the Bellman objective before optimizing a neural loss.	The original recursive risk operator, continuation, feasible actor, and complete-policy objective define the problem. A residual training penalty is a computational choice whose relation to the decision requires a theorem.	A training objective under another model or information set cannot establish improvement in this original objective.

Historical requirement	Treatment in the current NBO article and supplement	Remaining boundary
Mixed reward/residual penalties can select the wrong solution.	The original counterexamples and corrected objective discussion remain in the development record. Original-policy qualification uses outward Bellman supports and policy costs.	Archiving the counterexample does not validate the disproved mixed-objective claim.
An L^2 residual need not select the correct viscosity solution.	The supplement retains the counterexample and separates finite-horizon Bellman verification from PDE/BSDE residual statements.	Comparison, solution class, boundaries, and stability must be established for every claimed PDE extension.
Merton and constrained homothetic benchmarks contained incorrect formulas.	Corrected analytic identities, parameters, and boundary solutions remain the benchmark truth; invalid curves and values are not reused as active validation.	Reproducing an old erroneous script is historical evidence, not a successful analytic benchmark.
Source-constructed comparative curves were not independent experiments.	Their source and correction history remain visible. Active numerical claims require independently generated observations and named execution receipts.	The constructed DDPG curve cannot return as empirical competitive evidence.
Nonlinear recursive utility has domain and normalization requirements.	The Brownian Markov recursive-utility bridge states its own well-posedness and verification assumptions. Cardinal normalization and admissibility remain explicit.	This subcase does not solve general NDU, coupled FBSDEs, temporal-self equilibria, or games.
Gradients depend on the intended computational graph.	The current action certificate uses original global supports; the tail density is frozen when forming the all-action minorant. The deployed rounded actor is not silently differentiated as a smooth optimizer.	Stochastic trace bias, actor/critic graph choices, and training convergence require their separate assumptions and checks.
Existence, uniqueness, boundaries, and constraints must be explicit.	The original finite-horizon model uses compact feasible actions, original continuity and curvature, stated terminal policy, and constrained scalar minimizations.	No global conclusion about arbitrary neural objectives, fixed-step Adam, or arbitrary nonlinear transitions is inferred.

Historical requirement	Treatment in the current NBO article and supplement	Remaining boundary
Error control must align with actions.	The retained directional support, tail-switch bounds, completed-incumbent descent, remaining-gain stopping, and new active-tail/coupled certificates connect information to original policy loss.	Prospectively validated action-aware acquisition and its complete-work benefit remain separate empirical obligations.
Information access must be equal.	Formula access, exact values, uncertain outward values, derivatives, stored features, and training observations are distinct information models. The new exact-point theorem pays for tail localization.	Equal feature geometry available to a conventional hinge cannot establish neural exclusivity.
Conventional winners and failed runs are scientific evidence.	R49 FVI policy-cost wins, R62 adverse pure-ReLU comparisons, R67 vector work, R96 zero-crossing population intervals, and R97 quadratic/span negatives remain preserved with identities.	They cannot be erased by selective replacement, changed evaluation criteria, or a unrelated newer experiment.
Scaling must include the recursive workload.	Product points, descendant continuation evaluations, terminal queries, model acquisition, memory, rejected candidates, and fallback belong in total work.	Small kernel checks do not complete a learned long-horizon/control/innovation tournament.
Statistical claims require the actual frozen actors and independent sample.	Episode ledgers preserve source, actor, sample, protocol, and execution status. Deterministic catalogues remain deterministic.	R96, R98, R115, and other separately committed episodes cannot supply interchangeable population evidence.
Policy improvement must compare complete original policies.	The retained completed-incumbent rule accepts a new action only on a strictly negative certified original-policy contrast; the new common-density certificate charges both tail defects and $\pm\beta\varepsilon_1$.	A smaller continuation norm or a better certificate for the same action is not itself a lower policy cost.

Historical requirement	Treatment in the current NBO article and supplement	Remaining boundary
Reproduction must cover acquisition and executable closure.	Current independent fitting reconstruction, exact/rational checks where applicable, source support, and source-bound reproduction paths are included.	Shared-library limitations, missing historical dependencies, unsuccessful transports, and failed replay assertions remain disclosed.
The active paper needs focus without deleting history.	Current economic contribution and proofs are separated from the complete development companion and retention ledger.	A source/PDF hash certifies identity; it does not certify correctness, novelty, external acceptance, or an unexecuted experiment.

The original major-comment ledger remains the exact source for the wording and location of each historical requirement. The current response makes substantive progress where a theorem, corrected implementation, or executed comparison supports it; it leaves the unsatisfied empirical and theoretical obligations visible.

8 Completed R119 experimental record

The same original two-date investment model, continuous uniform law, state-dependent capacity, risk levels $1/2$ and $3/4$ and implemented terminal controller are used throughout. At the root, qualification requires original-policy regret at most $1/256$ and an original-policy interval width at most $1/8192$; the unchanged future-controller allowance is 2^{-20}. Each phase freshly acquires 256 labels per risk on the fixed training grid and evaluates all 144 declared states for eight representations. The three phases are separate, prospectively frozen development interventions on an already observed deterministic catalogue. B was designed after A and C after A and B. They are not independent unseen validation samples, and preceding data never substitute for contemporaneous controls.

Phase	Source prefix	Services	Primary decisions	Timing decisions	Exact record checks
A	1f33c266	3	6,912	10,752	444,053
B	357b2c56	5	11,520	17,920	1,031,933
C	095936f1	6	13,824	21,504	1,325,935

The full executable freezes are A: `1f33c2667e0dff89699c6d6a01d03eacfecf3f4e`; B: `357b2c56727cd57b9696cd6e07e115a35f5fdedb`; and C: `095936f1a7556d90d74809e63d75d9959d118360`. Every displayed decision and check count is from completed execution.

The respective attempt counts are 25161,45557 and 55043. Each phase completes seven sequential fresh timing processes. Its independent primitive implementation performs 22220 checks under that phase's source; these repeat the same declared deterministic validation contexts, rather than adding independent random observations. The native audits check exact binary-rational record inequalities and sequential policy identities; the continuum justification remains the theorem. In C, a separate reporter checks all 768 raw hashes and 384551 further conditions, and an additional 21974 checks reconstruct the paired original points and positive-mass packing geometry.

8.1 Preserved negative phases A and B

A reduces original risk-point queries from 654836 for full compiled qualification to 293844 for active-tail qualification, while the strong geometric service uses 223767. Active tail beats geometric in only one of 16 median-clock cells, by a ratio 0.9947 with only three of seven paired clocks lower. ReLU loses to both controls at both risks in every repeat. The exact 320-cell information experiment is separate: it records 242 narrower active-tail intervals,27 narrower whole-support intervals and 51 ties, including whole-support wins in 20 of 40 $\alpha = 0$ cells.

B's positive-tail masking reduces risk queries to 220341 but is slower than the original active-tail service in all 16 median cells. Its bounded two-level service uses 154217 queries but beats geometric in only two cells, ReLU and the equivalent hinge at risk 3/4. Summing all primary complete service observations gives 9.2482,7.2390,8.0358,8.6385 and 7.5186 seconds for compiled, geometric, active, masked and budgeted respectively. These adverse implementation results are preserved without retiming, changing the tolerance or omitting losing cells. All 5888 budgeted primary and timing rows obey the 30 outward and 16 boundary-payload caps, attaining both maxima. The full masked ladder and the two-level service both send the same 244 distinct primary proposals to the complete original fallback.

8.2 C: the same bounded information, packed into fewer calls

C uses exactly the same preliminary cut, risk-density defect, positive-mass nodes, two-level budget and original fallback. Only their compiler batching changes. All 5888 paired budgeted/fused decisions use the same original point coordinates and qualification exits. Fused makes 27 packed calls for the primary catalogue and 182 across all timing processes, containing 24193 and 35504 midpoint/endpoint payloads; the two outward defect endpoints remain separately paid. Total original risk-point queries remain 154217, exactly budgeted's count. Different outward endpoints occur in 600 primary and 2030 timing rows, so mathematical packing is not misreported as bitwise interval equality.

The fused service is faster in all 16 median cells against its contemporaneous budgeted and compiled controls, in 14 against active tail, and in ten against geometric. All six geometric losses remain in the table below. ReLU's fused/geometric median ratios are 0.8614 at risk 1/2 and 0.8494 at 3/4, with six and seven paired wins. Ordinary hinges also benefit. The degree-three ratios 0.9981 and 0.9999 are near-ties whose nominal median wins do not imply an operationally substantial advantage.

Risk	Representation	Geometric	Budgeted	Fused
1/2	Online	0.0767	0.1074	0.0794
1/2	Quadratic	0.0332	0.0621	0.0351
1/2	ReLU	0.0937	0.1093	0.0807
1/2	Hinge	0.0867	0.0998	0.0767
1/2	Spline	0.0319	0.0638	0.0358
1/2	Polynomial 2	0.1917	0.2194	0.2115
1/2	Polynomial 3	0.2520	0.2892	0.2516
1/2	Grid	0.1935	0.2266	0.2001
3/4	Online	0.0750	0.0857	0.0712
3/4	Quadratic	0.0334	0.0419	0.0296
3/4	ReLU	0.0936	0.0954	0.0795
3/4	Hinge	0.0872	0.0848	0.0720

Risk	Representation	Geometric	Budgeted	Fused
3/4	Spline	0.0331	0.0436	0.0315
3/4	Polynomial 2	0.1917	0.2106	0.1952
3/4	Polynomial 3	0.2403	0.2563	0.2402
3/4	Grid	0.1950	0.2098	0.1931

All entries are seconds, medians of the seven declared complete 32-state services; acquisition is charged separately. Full ranges, paired win counts and all six service columns remain in `new119/study_c119.tex` and the development tables. Across the entire primary tournament fused costs 6.5522 seconds, geometric 6.7726 and budgeted 6.8250. Fused is faster than geometric in 11 of 16 full 144-state cells and slower in five. These aggregates do not select a fastest route after observing each state.

8.3 Original policy value, conventional frontier and evidence limits

All three, ten and fifteen within-arm service-pair comparisons in A, B and C respectively return exactly the same complete original policies at every one of 2304 state–risk–arm combinations. Their economic contrasts are exactly zero. This is equality of policy functions on the declared current contexts with the same future controller, not a population inference or an equivalence conclusion from overlapping confidence intervals. The phases have 732,1220 and 1464 separately paid primary repair events respectively: the same 244 distinct proposals, serviced under three, five or six algorithms. No exact rational fallback is used in these completed campaigns.

Every distinct returned state/action receives a separately paid 1024-bin original-law reference under the unchanged future controller. Its integration partition is independent of the qualification mesh, while the terminal compiler is shared with production. It is therefore not a fully independent implementation of all 983 distinct policy references. The separate exact-rational primitive suite independently reconstructs 24 original reference cases and 72 certificates. The catalogue references give numerical objective and regret enclosures and strong-convexity action-distance bounds. The maximum recorded regret upper bound is 0.00285149 and the maximum distance upper bound 0.053400; they are valid bounds, not exact errors. No reference work is made free by deduplication: all actual reference construction and queries are retained in the paid record.

ReLU's catalogue-mean original cost exceeds quadratic's by $[1.79075, 1.87557] \times 10^{-5}$ at risk $1/2$ and $[1.80271, 1.88385] \times 10^{-5}$ at $3/4$. Statewise ReLU is cheaper at 43 and 39 states, so the full mean and statewise outcomes remain distinct. Against online control it has identical policy at 108 states and higher cost at 36 for each risk. The ReLU and ordinary-hinge policies, proposals and attempted certificates agree in all 96 paired C files and 4416 decisions; clock differences between their implementations are not a neural-exclusive representation property.

Charging fresh standalone acquisition, actual loading and complete 144-state service, quadratic with geometric costs 0.1485 seconds at risk $1/2$, versus 0.3407 for ReLU with fused. At $3/4$, quadratic with fused costs 0.1417, versus 0.3131 for ReLU with fused. The conventional quadratic therefore has both the lower full-workload resource cost and the lower mean original economic cost. C supplies a shared computational improvement in the original decision service; it establishes no neural adoption region over that frontier and no new policy-value gain for unchanged policies.

The completed numerical evidence is source-bound in `results119/primary119/AUDIT119.json`, `results119/primary119b/AUDIT119B.json` and `results119/primary119c/AUDIT119C.json`, with corresponding protocols, independent checks, raw attempts and references. The full A/B and C accounts appear in `new119/R119_STUDY_MAIN.tex`, `new119/R119_STUDY_DEVELOPMENT.tex`,

new119/study_c119.tex and new119/development_c119.tex. The phase-A formatting overlap, the pre-science B byte-transport rejection and the failed C reporting attempt are retained with their actual status. None changes a scientific protocol after its source freeze.

The unchanged least-squares fit does not complete action-aware learning; no population confidence interval, calibrated deployment price or learned multi-horizon/control/innovation tournament is supplied by these deterministic service experiments. Those requirements remain research objectives alongside the now-proved risk-uniform information rate, original-policy certificate and bounded implementation.

8.4 Independent validation D of every delivered original-policy key

We have extended the separately implemented original-value validation from the earlier 24-context Fraction suite to every realized policy key in phase C. The complete protocol, exact key catalogue and implementation were committed and pushed at cafd38aedc5bcbddf5aafee2481fbf5e9a267b ab before D evaluation. The runtime bound all 49 source files before its first economic calculation. The fixed resolution of 64 whole-support cells and 32 rational terminal steps was chosen after the known A/B/C outcomes. D is an exhaustive validation of that known catalogue. It changes no A/B/C action, fitted actor, primary observation or competitive clock. Its newly computed rational-node records belong to D alone and are not a new population sample.

The implementation imports only the unchanged standard-library Fraction economic primitive and reconstructs the original two-date investment model. It never imports the production terminal compiler. The exact stored binary64 states and actions are retained, without rounding them back to an intended grid. All 983 distinct state/action/risk keys—492 at $\alpha = 1/2$ and 491 at $\alpha = 3/4$—were included. The 13,824 original C records cover eight arms, six services and 144 states at each risk.

Enclosure construction and its precise scope. At each of the 129 rational midpoint/endpoint nodes, a separate terminal optimization calculation gives $\ell_k \le V_1(y_k) \le u_k$ from the original primitive values, exact continuous-risk density and a global constrained quadratic minorant. The squared aggregate hinge is constant in the original zero-aggregate innovation, so the terminal shock polynomial has its exact coefficient five. No generic three-point interpolation assertion is used. For every common threshold η, convexity of $(V_1(F(x,a,z)) - \eta)_+$ gives a midpoint lower expectation and an upper expectation from both endpoints of every cell. Interior endpoints occur twice in the latter law; they are not deduplicated. Taking the AVaR infimum over that common threshold yields the original continuous-risk enclosure.

Writing the resulting fixed-action bounds as $[L_Q, U_Q]$, the complete unchanged future policy satisfies

$$L_Q \le Q_x(a) \le U_Q, \qquad L_Q \le J_x(a) \le U_Q + \frac{15}{16}\,2^{-20}.$$

The final allowance is added once and uses the proved uniform performance bound of the unchanged future controller. D does not directly reconstruct the controller's 40-step floating implementation, independently select the date-zero action, or compute a new global date-zero optimum, regret or action-distance certificate. Its second-implementation guarantee is for the original value enclosure of the delivered actions. The full proof is supplied in new119/independent_validation119_proof.tex.

Executed coverage and all economic comparisons. All 983 keys and 126,807 terminal contexts completed, with 652,774 recorded checks and no failed key. The execution and external launcher both completed successfully, with exit code zero. Every fixed key and all 126,807 terminal contexts

are retained with their primitive traces; no unfavorable key or sign was removed. The separately executed completed-record reporter passed 590,818 read-only checks, including all saved terminal nodes and exact reaggregation of every mean, contrast and service identity. The output contains all 96 arm/service/risk mean intervals, all 56 arm-mean contrasts and all 240 within-arm service-pair identity cells. The mean always weights the same 144 original states equally. With the fixed left-minus-right arm ordering, the 56 contrasts comprise 32 strictly negative, 20 strictly positive, 2 exact-zero and 2 unresolved intervals. The two unresolved comparisons are quadratic versus the degree-two polynomial, one at each risk; the two exact zeros are ReLU versus the equivalent conventional hinge.

All 240 service-pair cells contain identical complete policies at every one of their 144 states, giving exact zero mean differences rather than an inference from overlapping intervals. At a state where two arms return the same complete policy, their difference is zero by the defining economic objective. Otherwise, the independent value intervals are subtracted before averaging. These are deterministic bounds, without a population confidence interpretation or a required favorable sign. The independently enclosed ReLU-minus-quadratic mean differences are:

Risk α	Original mean cost difference	Independent conclusion
$1/2$	$[1.7436, 1.9224] \times 10^{-5}$	strictly higher ReLU cost
$3/4$	$[1.7533, 1.9360] \times 10^{-5}$	strictly higher ReLU cost

All 56 comparisons, including unresolved signs, remain in the development tables. The earlier fine 1,024-cell references are preserved and explicitly retain their shared-terminal-compiler scope. D does not rewrite those fine endpoints or transfer its observations into any historical population study.

Compatibility and additional work. All 13,824 recorded service intervals and all 983 shared-compiler fine references overlap their independent enclosure. The full independent enclosure is contained in 10,300 service intervals and 0 fine references; the reverse inclusions occur in 3,524 and 983 cases. Conditional on validity of the independently proved enclosure, the 10,300 containing service intervals inherit value coverage. Reverse containment and overlap alone do not prove a finer recorded endpoint correct. In particular, a nonempty intersection cannot establish that either endpoint of the fine production interval is itself correct. The independent enclosure instead relies on the stated mathematical assumptions and the separate primitive implementation; overlap is an additional diagnostic.

The actual external process cost is 1697.331508 seconds, including 1697.251692 seconds measured inside the process and 1693.603548 CPU seconds; recorded peak resident memory is 83,152 KiB, and compressed primitive traces occupy 48,026,132 bytes. The separately executed completed-record audit costs a further 7.705036 elapsed and 7.538375 CPU seconds. This cost includes source/input binding, independent solves, trace creation and hashing, comparisons, logging and aggregation. It is additional author-side verification expense, separately reported from every unchanged A/B/C acquisition and service cost. The repeated competitive clocks and all historical negative outcomes remain intact.

The completed records are `results119/validation119d_attempt01/EXECUTION119D.json` and `results119/validation119d_attempt01/SUMMARY119D.json`, with the external command, output hash, elapsed time and return status in `results119/logs/validation119d_attempt01.json`. The manuscript and full numerical treatment are `new119/study_d119.tex` and `new119/development_d119.tex`. This validation improves the independent original-value evidence; it does not complete the outstanding action-aligned training theorem, independent learned-to-action solver, calibrated adoption application or neural-specific advantage.

9 Publication, custody and source-specific verification record

The author revision is confined to `revision/econometrica-nbo-r119-active-tail-author-revision-2026-10-18`. The remote reference audit at 2026-10-18 06:47:43 UTC verifies that all 359 pre-existing branch heads are unchanged; only this isolated revision branch has been added. Its recorded evidence commit `2665228a4ea23a2609f74e3e3aeae7d36d7e53b6` preserves the raw archives and their member/hash manifests. The exact check is `provenance/FINAL_BRANCH_PRESERVATION119.json`. Subsequent publication commits on that same new branch do not replace the earlier experimental source freezes.

The native diagnostic build starting 2026-10-18 06:46:10 UTC ran the four requested LaTeX documents through repeated PDF passes and BibTeX. Every recorded command exited successfully, and its final log-issue map is empty. The receipt `build_receipts119/20261018T064610.847294Z/BUILD_RECEIPT119.json` binds those exact input sources, command logs and four PDF hashes. It is explicitly a diagnostic build of the recorded inputs; this final response and additional historical companion are later author-text integrations. The final publication package therefore supplies a separate source/PDF manifest and actual remote workflow record. These external records identify the released commit, artifact hashes and observed CI status without requiring a PDF to contain its own output hash. A diagnostic build, mathematical check, economic experiment and remote publication check remain different evidence objects.

GitHub run 39101786083, "NBO R119 immutable R110 custody recovery," is independently verified as completed with conclusion success. It ran at commit `357b2c56727cd57b9696cd6e07e115a35f5fdedb` and its recorded completion update is 2026-10-18 06:00:08 UTC. Its successful scope is recovery and verification of the selected immutable R110 artifact. The original manuscript identity remains `4ba2bbaa11b5fa59f83705b28b35bf82a63ca169`. The custody receipt does not assert that the final revised PDFs or a new historical economic experiment ran in that workflow.

The complete historical byte/read union now covers 66471 unique Git blobs reachable from all 359 original branch references, totaling 17202527633 bytes. It includes all 10358 initially missing objects, whose 15397774846 bytes were read and Git-verified by the local immutable-commit auditor; all nine initial HTTP502 attempts are preserved and the corresponding retries passed. The actual records are `provenance/historical/ALL_HISTORY_READ_CLOSURE119.json` and `provenance/historical/SUMMARY119.json`. Safe format interpretation, opaque binary custody, mathematical proof review and scientific reproduction are separate scopes. In particular, the historical syntax defects, truncated R118 transport and uninterpreted opaque payloads remain explicitly recorded. No success conclusion for remote workflow 39103777344 is inferred from this local closure.

Preserved historical-custody CI failure and its repair. The job-level evidence for historical-custody run 39103777344 supersedes the earlier observation of a queued run endpoint. The final saved job observation, checked at 14:31 and 14:37 UTC on 2026-10-18, records all sixteen jobs completed with failure at checkout; none started the immutable-blob audit. The earlier 07:21 UTC snapshot of twelve failures and four queued jobs remains preserved as the observation made at that time. The representative log identifies the cause: the job-wide `GIT_NO_LAZY_FETCH=1` setting prevented the declared sparse checkout from fetching a missing blob from the promisor remote. Both the job records and the decoded log remain in `provenance/historical/ci_attempt01/`.

Commit `cafd38aedc5bcbddf5aafee2481fbf5e9a267bab` confines this guard to the bounded audit command, after the explicit checkout. The source repair is separately recorded; it is not evidence that the repaired workflow has passed. Its separate run 39136070575 still had all sixteen

jobs queued in the direct job-endpoint observation at 14:46 UTC. The completed local historical byte audit and its preserved HTTP failures and retries are also distinct from this failed remote attempt.

The successful R110 custody-recovery run 39101786083 has its own immutable source identity and completed success receipt. Its scope is recovery and verification of the R110 artifact. Neither that success, the local historical audit, a numerical validation receipt, nor a workflow repair stands for the final R119 source and PDF build. The external publication receipt records the exact current source, actual output hashes and observed status of each final build or CI attempt without requiring a PDF to contain its own hash.

The completed scientific advances remain tied to the original Bellman objective. The full author response, supplement, development companions, protocols, exact source identities, numerical receipts and retention maps are provided together. Phase A and B failures, the six phase C geometric losses, the conventional economic winner and every retained scientific obligation remain available for assessment.